

Saiteja Venkateshwa Rao Dasari

Robotics Engineer | Controls Engineer | Autonomy Engineer

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SUMMARY

M.S. Robotics graduate with UAV integration and autonomy experience across ROS2, UAV GNC, fault-tolerant control, and sim-to-hardware testing; strong in Python, C/C++, and MATLAB/Simulink for controls, path planning, and real-time robotic systems.

TECHNICAL SKILLS

Programming/Tools: Python, C/C++, MATLAB/Simulink, Bash, Linux, Git, SolidWorks.

Robotics/Simulation: ROS1/ROS2, Gazebo, MuJoCo, CrazySim SITL, RViz, tf2, URDF/xacro, ArduPilot, Mission Planner, cflib.

Controls/GNC/Test: PID, closed-loop control, state estimation, sensor fusion, FDI, control allocation, controller reconfiguration, SITL, flight-log analysis.

Autonomy/Perception: A*, Hungarian algorithm, path planning, costmaps, occupancy grids, collision avoidance, OpenCV, YOLOv8, ArUco, pose estimation.

EXPERIENCE

Honeywell Aerospace

Aug 2025 – Apr 2026

Technical Lead — SkySpeak AI (ASU Collaborative Research)

Tempe, AZ

- Directed a 3-engineer team with Honeywell/ASU stakeholders to deliver an ATC/pilot simulator with LLM agents, guardrails, and a 300 ms LiveKit/Deepgram/ElevenLabs voice pipeline.
- Engineered Python voice analytics with YIN pitch detection, VAD, z-score calibration, and abstention scoring (<0.60); launched a CBTA dashboard for 20+ stakeholder reviews and seed-grant support.

Arizona State University

Jan 2025 – May 2026

Graduate Teaching Assistant — MATLAB Programming, Instrumentation & Controls Lab

Tempe, AZ

- Mentored 90+ students across MATLAB, instrumentation, and controls labs; troubleshoot code/hardware setups, led reviews, and reinforced PID/control-system concepts.

Marut Drones (IIIT-Hyderabad)

Aug 2023 – May 2024

Production Engineering Intern — UAV Systems Integration

Hyderabad, India

- Assembled and verified UAV subsystems (ESCs, flight controllers, GPS, avionics, propulsion), raising production efficiency by 15% across multirotor platforms.
- Tuned ArduPilot/Mission Planner PID parameters from flight logs across 3 UAV variants; root-caused stability issues and improved flight stability by $\sim 10\%$.

PROJECTS

Fault-Tolerant Quadrotor Control [GitHub] – Crazyflie 2.1, MuJoCo, CrazySim SITL

Jan 2026 – May 2026

- Derived and validated closed-form 3-motor allocation for all 4 single-motor failures, reaching 0° attitude error across 24 test conditions; root-caused attitude loss to motor saturation and added a thrust clamp that cut max roll from 145° to 7.2° on hardware.
- Built a MuJoCo rigid-body simulation modeling gyroscopic coupling and validated sim-to-real transfer across PID, INDI, and gyroscopic-compensation controllers, testing 9 allocation methods on hardware.

Multi-Robot Task Allocation & Navigation [GitHub] – ROS2, Gazebo, Python

Dec 2025

- Developed a centralized ROS2/Gazebo autonomy stack for 5 TurtleBots with from-scratch Hungarian assignment, a custom A* planner (ROS2 service), and an inflation costmap; attained 99% mission success.
- Built a path cache keyed on start pose and task ID to align allocated and executed routes, cutting redundant A* re-planning.

Autonomous Drone Landing on Moving Platform [GitHub] – MATLAB/Simulink, PID

May 2025

- Built a Parrot Mambo landing system with HSV segmentation, centroid tracking, velocity-sync logic, and staged PID descent to align with a moving 11×11 in platform; achieved <5 cm lateral error, ~ 4 s landing, and $>90\%$ success across varied platform speeds.

Crop & Weed Detection [Portfolio] – ROSMaster X3, ROS, YOLOv8, OpenCV

May 2025

- Deployed real-time YOLOv8 crop/weed detection on ROSMaster X3 using 3,000 samples; connected ROS inference and visualization nodes and reached 97% accuracy at 12 FPS.

Maze Solving & Path Planning [GitHub] – myCobot Pro 600, OpenCV, A*, IK

Dec 2024

- Built an OpenCV vision pipeline into a 34×34 occupancy grid, then computed A* paths and inverse kinematics (MATLAB RigidBodyTree) to execute waypoint trajectories on a physical 6-DOF myCobot Pro 600 arm.

EDUCATION

Arizona State University

Aug 2024 – May 2026

M.S. Robotics and Autonomous Systems, Systems Engineering | GPA: 3.93/4.0

Tempe, AZ

Coursework: Aerial Robotics, Mechatronics, Multi-Robot Systems, Controls and Systems.

Chaitanya Bharathi Institute of Technology

Aug 2020 – May 2024

B.E. Electrical and Electronics Engineering

Hyderabad, India

CERTIFICATIONS & PUBLICATION

Universal Robots e-Series Core/Pro | MathWorks AI | Udemy ROS/ROS2 Gazebo | [SCADA-Based Substation Control Panel](#)